

**Instructions to the Students:**

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q.1. Objective Type Questions (Answer All the Questions)**(10×2) = 20**

- i) Write note on ozonolysis reaction.
- ii) Enlist qualitative test for aldehyde.
- iii) Explain Inductive effect.
- iv) What is saytzeff's Rule, give its reaction.
- v) Draw down the structure of I) 1-Methyl-3-Cyclohexanol II) 3-Hexanone
- vi) Define functional isomerism with example.
- vii) Explain Basicity of Amine.
- viii) Define the term nucleophile and leaving group.
- ix) Describe any two qualitative tests of carboxylic acid.
- x) Predict the product as per

**Q.2. Long Answers (Answer 2 out of 3)****(10×2) = 20**

- i) What is elimination reaction? Explain reaction, mechanism kinetics & order of Reactivity of E1 & E2 Reaction.
- ii) Why Aldehydes & ketones are susceptible for nucleophilic addition reaction? Explain the reaction & mechanism of Aldol condensation & crossed aldol condensation.
- iii) Explain the Reaction, mechanism & stereo chemistry of SN1 Reaction.

Q.3. Short Answers (Answer 7 out of 9)**(5×7) = 35**

- i) Write down the IUPAC Nomenclature rule for organic compound.
- ii) What is carboxylic acid. Explain effect of substituents on acidity of carboxylic acid.
- iii) Describe the Perkin Condensation Reaction.
- iv) Define alkyl Halide & give it different chemical Reaction shown by alkyl halide.
- v) Explain Qualitative tests for Amines?
- vi) Draw the structures and uses for following organic compounds a) Chlorobutanol b) Acetyl salicylic Acid.
- vii) Describe Halogenation of Alkane with examples.
- viii) Explain Diel – Alder reaction.
- ix) Write a note on Isomerism of organic compounds.

*** All The Best ***